TSM 098

Homework 7

Coordinate graphs and linear systems

Topics 30-31, 33 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Points and lines

1. Plot (-2, 3), (0, -1), and (4, 0). State the quadrant or axis of each point.

2. Graph using intercepts and a third point; check all three points in the equation.

$2x + 3y = 6$

3. Graph x = -2 and y = 3. Give their intersection and check it in both equations.

4. Graph both equations using points. Estimate the intersection, then check it exactly.

$x + y = 5, \qquad x - y = 1$

Algebraic systems

5. Solve by substitution. Check both equations.

$y = 2x + 1, \qquad x + y = 7$

6. Solve by elimination. Check both equations.

$2x + 3y = 13, \qquad 2x - y = 1$

7. Solve by elimination or substitution. Show any preparation needed.

$3x + 2y = 4, \qquad 5x - 2y = 12$

8. Solve and describe the full solution set.

$x + 2y = 6, \qquad 2x + 4y = 12$

9. Solve and describe the full solution set.

$x + 2y = 6, \qquad 2x + 4y = 10$

10. Does (2, 1) solve the system? Explain why checking just one equation is insufficient.

$x + y = 3, \qquad 2x - y = 5$

11. Choose substitution or elimination, solve, and explain why your method is efficient.

$x = 3y - 2, \qquad 2x + y = 10$